

Data base management systems

Minor project

Student Performance Analyzer using Python and MySQL

Submitted By:

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CSE

Submitted to

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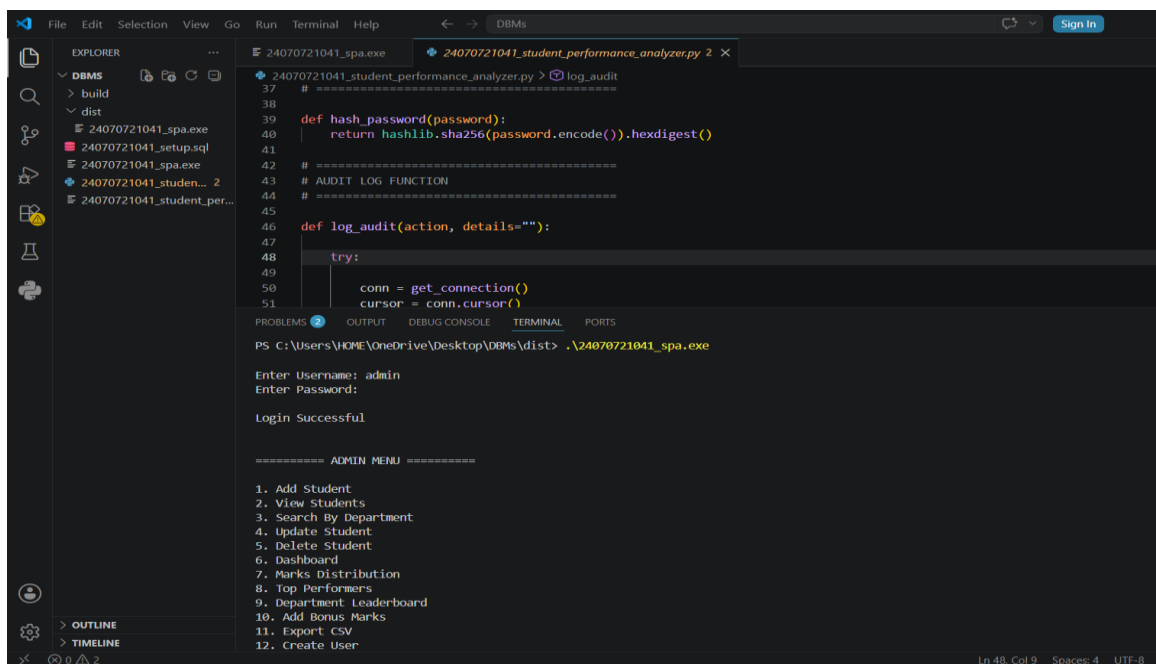
Assistant Professor

DATABASE MANAGEMENT SYSTEMS LAB MINI PROJECT REPORT

STUDENT PERFORMANCE ANALYZER USING PYTHON AND MYSQL

Introduction

The Student Performance Analyzer is a database-driven mini project developed using Python and MySQL. The main objective of the project is to manage student academic records securely and efficiently using role-based access control. The system supports three types of users: Admin, Faculty, and Student, each having different permissions and functionalities. The application performs CRUD operations on student records, generates analytics reports, exports high-performing students into CSV format, and maintains audit logs for tracking important activities.



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24070721041_student_performance_analyzer.py
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39 def hash_password(password):
40     return hashlib.sha256(password.encode()).hexdigest()
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42 # =====
43 # AUDIT LOG FUNCTION
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46 def log_audit(action, details=""):
47     try:
48         conn = get_connection()
49         cursor = conn.cursor()
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records but cannot delete students or manage users. Students can only view their own records and leaderboard information.

3. Password Security

Passwords are not stored directly in the database. SHA-256 hashing was used to securely store password hashes. During login, entered passwords are hashed and compared with stored values to improve security.

4. Parameterized Queries

Parameterized SQL queries were used throughout the project to prevent SQL injection attacks and improve application security.

5. Audit Logging and Analytics

All INSERT, UPDATE, DELETE, and LOGIN operations are recorded in the audit_log table for monitoring and security purposes. The project also includes analytics features such as average marks, highest and lowest marks, grade distribution, department leaderboard, and top performers using SQL functions like AVG(), MAX(), MIN(), GROUP BY, ORDER BY, and CASE.

Challenges Faced and Solutions

1. Python–MySQL Connection Error

Solution: Configured correct MySQL credentials and installed mysql-connector-python properly.

2. Password Authentication Failure

Solution: Implemented SHA-256 hashing in both Python and MySQL for secure password verification.

3. Role-Based Access Restriction

Solution: Created separate menu functions for Admin, Faculty, and Student to control permissions secure

Use of AI

AI tools were used for understanding Python-MySQL integration, debugging connectivity issues, improving SQL query structure, generating code explanations, and preparing documentation and viva questions. All generated code was manually tested, modified, and integrated according to the project requirements.

Conclusion

The Student Performance Analyzer successfully demonstrates the implementation of DBMS concepts using Python and MySQL. The project includes secure authentication, role-based access control, CRUD operations, analytics, audit logging, and CSV report generation. The project helped in understanding practical applications of database management systems, SQL queries, and Python database connectivity in real-world applications.